

# IS EVALUATION AWARENESS JUST FORMAT SENSITIVITY? LIMITATIONS OF PROBE-BASED EVIDENCE UNDER CONTROLLED PROMPT STRUCTURE

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## ABSTRACT

Prior work uses linear probes on benchmark prompts as evidence of *evaluation awareness* in large language models. Because evaluation context is typically entangled with benchmark format and genre, it is unclear whether probe-based signals reflect context or surface structure. We test whether these signals persist under partial control of prompt format using a controlled  $2 \times 2$  dataset and diagnostic rewrites. We find that probes primarily track benchmark-canonical structure and fail to generalize to free-form prompts independent of linguistic style. Thus, standard probe-based methodologies do not reliably disentangle evaluation context from structural artifacts, limiting the evidential strength of existing results.

## 1 INTRODUCTION

Recent work introduces the notion of *evaluation awareness* in large language models (LLMs): the ability to distinguish evaluation-time prompts from deployment-like usage and potentially modulate behavior accordingly (Needham et al., 2025; Chaudhary et al., 2025). This phenomenon has implications for benchmark validity and safety monitoring, as evaluation-time behavior may not reflect real-world deployment (Hua et al., 2026).

Empirical evidence for evaluation awareness largely relies on *probe-based analyses*, where linear classifiers trained on frozen activations distinguish evaluation from deployment prompts (Nguyen et al., 2025). However, probe accuracy is not diagnostic: classifiers may exploit any feature correlated with labels. In practice, evaluation prompts are tightly coupled with benchmark-canonical formats and neutral genres, while deployment prompts are more conversational (Needham et al., 2025). This coupling raises the concern that probes track surface structure rather than evaluation context itself (Figure 1a).

We ask a methodological question: *Do probe-based signals attributed to evaluation awareness remain informative once prompt format and genre are partially controlled?* Rather than testing whether evaluation awareness exists, we assess whether standard probe-based methods can isolate evaluation context under stricter controls.

Prior work reports cross-dataset generalization of evaluation-awareness probes, but existing datasets rarely break the correlation between evaluation context and benchmark-canonical format. We introduce controlled format-transplant and genre controls showing that, under the standard single-contrast training setup, probe signals largely track benchmark structure rather than context.

We make three contributions:

- **Controlled  $2 \times 2$  design.** We construct datasets and diagnostic rewrites that independently vary context, genre, and format under matched prompt length (Figure 1b).
- **Probe generalization fails under format-transplant interventions.** Linear probes trained on benchmark contrasts fail to generalize to free-form prompts, even when content and style are preserved.

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- **Methodological implications.** Probe-based evaluation-awareness diagnostics are highly format-sensitive, limiting their use as structure-invariant safety tools.

## 2 RELATED WORK

Several recent studies examine whether LLMs internally distinguish evaluation from deployment settings. Needham et al. (2025) show that state-of-the-art models can classify transcripts as evaluation or deployment with high accuracy ( $AUC \approx 0.83$ , compared to a human upper bound of  $\approx 0.92$ ), suggesting that prompts contain systematic cues associated with evaluation contexts.

Nguyen et al. (2025) probe frozen internal activations using linear classifiers and report generalization across held-out datasets. They observe that safety-related benchmarks are consistently flagged as evaluation prompts even without explicit training, indicating that benchmark structure itself may act as a strong cue. Relatedly, Hua et al. (2026) train a synthetic “organism” model to exhibit evaluation-aware behavior and identify an activation direction whose removal suppresses such behavior, demonstrating that evaluation-linked representations can be induced and manipulated.

Chaudhary et al. (2025) study how probe-based evaluation awareness scales with model size, reporting a power-law relationship across open-weight models ranging from 270M to 70B parameters. Together, these results suggest that probe-detectable signals associated with evaluation settings are widespread and scale systematically.

Our work differs by testing whether such probe-based signals persist under stricter controls on prompt format and genre. Rather than asking whether evaluation awareness exists, we assess whether standard probe-based diagnostics isolate evaluation context or instead reflect sensitivity to benchmark-canonical structure.

## 3 SETUP AND METHODS

We investigate whether probe-based evaluation awareness is a genuine response to usage context or an artifact of sensitivity to prompt format. Unlike prior work that conflates these dimensions, our design treats *Context* (Evaluation vs. Deployment) and *Format* (Benchmark vs. Casual) as independent variables.

### 3.1 THE 2X2 CONTROLLED DESIGN

We construct a dataset matrix covering four quadrants to isolate the effects of structure. We define our variables as follows:

- **Context:** The underlying intent. *Evaluation* tests capabilities; *Deployment* assists a user.
- **Format:** The surface structure. *Benchmark* implies rigid templates (e.g., MCQ); *Casual* implies free-form inputs.

**Dataset Definitions.** **Bench-Eval** consists of canonical prompts from MMLU, ARC, and GSM8K. **Casual-Deploy** uses deployment-style user prompts sampled from ShareGPT (restricted to the first

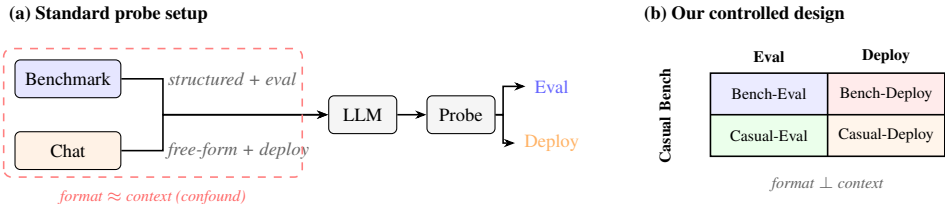


Figure 1: Overview of our approach. (a) Standard probes train on benchmark vs. chat prompts, where format and context are confounded. (b) Our  $2 \times 2$  design crosses format and context independently, enabling isolation of each factor.

Table 1: The  $2 \times 2$  Dataset Matrix. We cross usage context with prompt format to disentangle structural cues from evaluation intent.

| Format                           | Evaluation Context                                                      | Deployment Context                                                         |
|----------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Benchmark</b><br>(Structured) | <b>Bench-Eval</b><br>(MMLU, ARC, GSM8K)<br><i>Standard Evaluation</i>   | <b>Bench-Deploy</b><br>(ShareGPT rewritten to<br>benchmark format)         |
| <b>Casual</b><br>(Free-form)     | <b>Casual-Eval</b><br>(MuTual, 1st turn)<br><i>Conversational Proxy</i> | <b>Casual-Deploy</b><br>(ShareGPT, 1st turn)<br><i>Standard Deployment</i> |

turn). **Casual-Eval** acts as a conversational evaluation proxy, using the first turn of the MuTual dataset. Finally, **Bench-Deploy** serves as an adversarial control: we route ShareGPT queries to compatible tasks and rewrite them into strict benchmark formats using DeepSeek-V3 (see Appendix A.6 for prompts).

### 3.2 EXPERIMENTAL SETUP

To ensure a clean experiment, we apply strict controls:

- **Length Control:** To prevent probes from relying on length heuristics, we applied *histogram matching* to align the sample length distribution of *Casual-Deploy* with *Bench-Eval*. Figure 2 in Appendix A.2 illustrates the resulting distributions.
- **Leakage Removal:** We explicitly audited rewrites to remove artifacts such as "Answer:" fields.
- **Model and Probes:** We extract frozen residual stream activations (last token) from Llama-3.1-8B-Instruct across middle-to-late layers (indices 8–31, 0-indexed). For each layer, we train a Logistic Regression probe using 5-fold cross-validation. Detailed hyperparameters are provided in Appendix A.4.

### 3.3 TRAINING CONFIGURATIONS

We compare two training paradigms with distinct evaluation protocols:

- **Standard Training (Single-Contrast):** Trained on the typical confound (*Bench-Eval* vs. *Casual-Deploy*). We do not use a validation split here, as models trivially achieve near 100% accuracy via format shortcuts. Instead, we evaluate *zero-shot* on held-out, format-shifted datasets (OOD evaluation) to diagnose generalization failure.
- **Paired Training (Decorrelated):** Trained on pooled data (*Bench-Eval*  $\cup$  *Casual-Eval* vs. *Bench-Deploy*  $\cup$  *Casual-Deploy*). Here, we use a standard **80/20 train/test split** to demonstrate that the model can learn a generalizable context signal (high test accuracy) when format is decorrelated from the label.

## 4 OBSERVED OUTCOME

Our experiments demonstrate that standard probe-based evidence for evaluation awareness is fragile. We present results for Layer 24 (consistent trends across all layers are reported in Appendix 4).

**The Format Trap.** As shown in Table 2 (top row), standard probes fail to generalize. Deployment prompts rewritten into benchmark formats (*Bench-Deploy*) are misclassified as "Evaluation" (94.5%). The probe detects the structural format (e.g., MCQ), not the context. To isolate **Genre** (linguistic register), we tested deployment prompts rewritten into a formal "Academic Tone" without structural changes. This had negligible effect ( $\approx 0\%$  classified as Eval), confirming that probes are sensitive to rigid templates rather than linguistic style.